Engine Model(s): PT6A

73-10-06

FUEL DRAIN VALVES - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-061	Cloth, Abrasive Coated
PWC06-003	Compound, Anti-galling
PWC06-015	Beeswax
PWC06-032A	Compound, Anti-galling
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

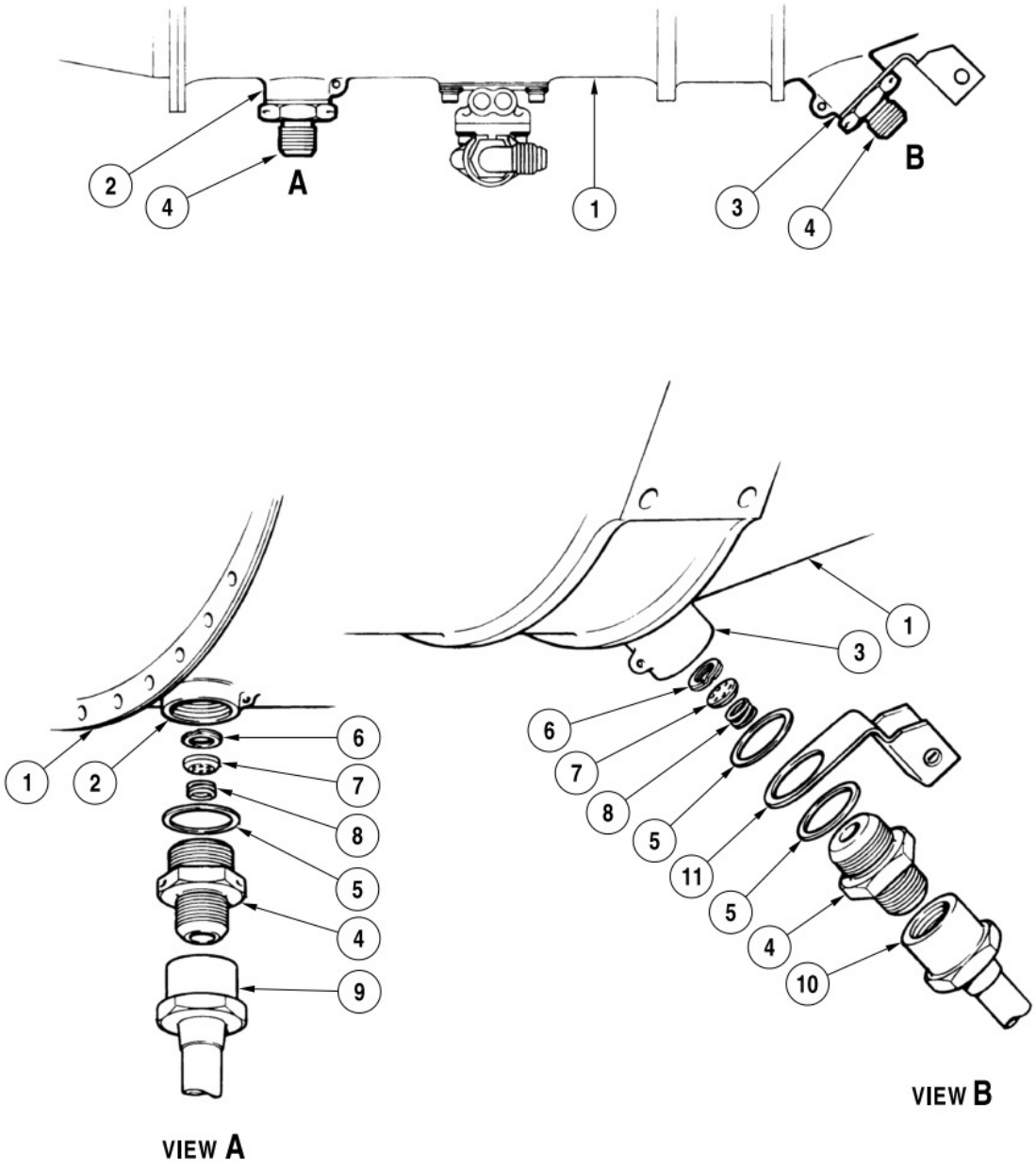
Not Applicable

5. Removal/Installation

A. Removal of Fuel Drain Valves (Ref. Fig. 201)

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Figure 201 Removal/Installation of Fuel Drain Valves



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1. Gas Generator Case
2. Front Drain Valve Boss
3. Rear Drain Valve Boss
4. Drain Valve Adapter
5. Gasket
6. Retaining Ring
7. Drain Valve
8. Spring
9. Front Drain Tube
10. Rear Drain Tube
11. Support Bracket

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(1) If the drain valve is being removed with the engine installed, disconnect the appropriate airframe drain line (9 or 10) from the adapter (4) and install a suitable blanking cap on the line connection.

(2) Disconnect and remove the primary and secondary fuel lines (Ref. 73-10-03).

(3) Unscrew each adapter (4) from the boss (2) on the gas generator case (1) and remove the gasket (5) from the adapter. At the rear valve (View B), remove the additional gasket (5) and support bracket (11).

(4) Remove the retaining ring (6) from the groove in the bore of each adapter (4) and withdraw the valve (7) and spring (8).

B. Installation of Fuel Drain Valves (Ref. Fig. 201)

(1) Insert the spring (8) into the bore of the adapter (4).

(2) Position the valve (7), flat side out, over the spring (8). Press the valve into the bore of the adapter (4) and install the retaining ring (6).

(3) For the front drain valve (View A), assemble the gasket (5) on the large diameter end of the adapter (4); apply a thin film of anti-galling compound (PWC06-003) or (PWC06-032A) or (PWC06-015) to the threads at the large diameter end of the adapter (4) and screw the adapter into the boss (2) on the gas generator case (1). Tighten the adapter 450 to 550 lb.in., but do not fasten with lockwire.

(4) For the rear drain valve (View B), assemble two gaskets (5) and the support bracket (11) on the large diameter end of the adapter (4); apply a thin film of anti-galling compound (PWC06-003) or (PWC06-032A) or (PWC06-015) to the threads at the large diameter end of the adapter (4) and screw the adapter into the boss (3) on the gas generator case (1). Tighten the adapter 450 to 550 lb.in., but do not fasten with lockwire.

NOTE: You must do the final torque and lockwire of the adapters (4) after an engine wet motoring run (Ref. Adjustment/Test).

(5) Install the primary and secondary fuel lines (Ref. 73-10-03).

(6) If installation of the drain valves has been carried out with the engine installed, remove the blanking caps from the line connections and connect the drain lines (9 and 10) to the adapters. Tighten the coupling nuts (Ref. Aircraft Maintenance Manual).

6. Cleaning/Painting

Clean all parts with solvent (PWC11-027) or (PWC11-031) and dry with filtered compressed air and/or clean, lint-free cloths.

7. Inspection/Check

A. General

(1) Examine the valve surface for nicks, gouges, scores, scratches and burrs. Particular attention must be given to the sealing surface around the counterbore in the spring side of the valve.

(2) Check for blockage in the six drain holes of the valve. Holes which are blocked may be

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cleared by passing through a 0.042 in. (No. 58) drill by hand.

(3) Examine the adapter for thread damage.

(4) Check for blockage in the centerbore of the adapter. Blockage may be cleared by passing through a 0.246 in. (letter D) drill by hand.

(5) Examine the large bore of the adapter for surface defects, especially on the sealing surface of projection.

(6) Examine the support for surface damage. A bent or distorted bracket should be replaced.

B. Drain Valve Spring Inspection Requirements

(1) Check the drain valve spring for distortion. The spring must pass freely over a 0.530 in. diameter gage rod and through a 0.600 in. diameter gage tube. Reject if distorted.

(2) Check the compression loading of the spring. The load required to compress the spring to 0.250 in. length should be 4 oz. to 8 oz. Reject the spring if the load is not within the limits.

(3) Examine the spring for uniformity of the coil spacing. Check that the surfaces are free from imperfections such as pits, scratches, nicks and other defects which may be detrimental to the fatigue resistance of the spring.

8. Approved Repairs

A. Procedure

(1) Repair of the component parts of the drain valves is limited to local blending of minor surface defects such as nicks, burrs, scratches, scores and gouges.

(2) Clean up these defects by blending with a fine stone and/or polishing with crocus cloth (PWC05-061).

(3) Remove all sharp edges and high spots.

(4) Thread damage on the adapter may be cleaned up with a suitable Swiss file or by chasing with an appropriately sized die.

(5) Prior to the installation of the adapters, make sure that the threads in the bosses on the gas generator case are not damaged; repair as necessary.

9. Adjustment/Test

A. Operation

(1) During the next engine test, do a wet motoring run (Ref. 71-00-00) and check the operation of the drain valves. The proper functioning of the valves will be indicated by the fuel flow from the drain lines.

(2) If fuel does not flow from one or both drain valves, remove the suspect valve, clean and reinstall. Repeat the wet motoring run and check the operation of the valves.

B. Procedure after Test (Ref. Fig. 201)

(1) Disconnect the drain lines (9 and 10) from the adapter (4).

(2) Loosen the adapters (4) two or three turns, and retighten 450 to 550 lb.in.

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(3) Lockwire the adapter (4) to the lug on the appropriate boss (2 or 3) on the gas generator case (1) and reconnect the drain lines (9 and 10).

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